

Project Description Proposals in the Area of Scientific Library Services and Information Systems

LIS Funding Programme "Information Infrastructures for Research Data"

World Heat Flow Database Project

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1. Starting point and preliminary work

The WHFD project in a nutshell – The *World Heat Flow Database* project aims at developing a modern research data infrastructure for global terrestrial heat-flow data, enabling the geoscientific community to efficiently access the results of 100 years of global heat-flow research and data collection. Thereby, we aim to provide a **national contribution to an international data collection and develop the new heat-flow research data infrastructure** that allows **extended interpretations based on modern data analysis and visualization tools**. At present, the data quality is heterogeneous on both national and international scale and without exception poor. Main reasons are rudimentary or even missing metadata and data documentation, especially as most data derives from scholarly literature and reports that primarily did not aim at providing comprehensive data descriptions and metadata. For Germany, a digital heat-flow database is not available. On the global scale, the available data collection is based on basic "agreements on descriptive elements" from 1976 that (1) do not sufficiently describe the data and (2) are not compliant with modern methods for heat flow determination. There is neither an online database nor a sustainable and fixed maintenance culture of existing data. Thus, from the national to the global scale, heat-flow related research faces the same fundamental challenges.

Within the proposed project, our activities **to close the national gap shall become the nucleus for the worldwide implementation** of a new global research data infrastructure for heat flow. Based on a sustainable database architecture, the new infrastructure will follow international metadata standards, will be equipped with visualisation features, data analytics, a data publication service with DOI, standard interfaces and web services to allow interoperability with other research infrastructures for the benefit of the international geoscientific community. To the research field of (Earth's) thermal-field analysis, a **revised and quality-controlled version of the existing heat-flow database** shall provide curated data in the spirit of FAIR and openness. Our long-standing expertise in the development of data information systems, heat-flow research, and our leading and initiative role in the activities of the international heat-flow community and commission enables us to tackle this challenge successfully.

Numerous world leading heat-flow experts, including the International Heat Flow Commission (IHFC) are supporting this project, honouring the integrated approach of data infrastructure development and will contribute to the collaborative data revision (→ see reference letters attached). On a global scale, we have recently initiated a Task Force of the International Lithosphere Program (ILP) that, aligned with the IHFC, will help to organize the global assessment of the historical heat flow data. Leading the TF (S.F.) will help us to attract the global lithospheric community for our assessment project. Within the 3 years of the project runtime, we expect to develop the new database as a **one-stop shop for heat-flow information and data** and to **revise all German heat-flow data and up to 60% of the globally available historical heat-flow data**. The complete revision is expected to be achieved within 2 years after the end of the DFG funding period. Following the initial DFG project funding, the new research data infrastructure will be operated and maintained as permanent service by the GFZ German Research Centre for Geosciences (Sections "Library and Information Services" and "Geoenergy").

What is heat flow – why is it important?

The terrestrial heat-flow density (simply called 'heat flow') is a measure of thermal energy flowing from the core towards the Earth's surface, it varies locally and is given as the rate of heat transferred across an area per unit time. Heat flow provides indispensable insights into the evolution of our planet and is paramount for the analysis of the Earth's temperature field across spatial scales and time domains. It improves our understanding of geodynamic processes and complex tectonic structures, the transient counter equilibrium of glacial cooling events, and the basic mantle heat flow entering the overlying crustal rocks. Therefore, heat-flow values are systematically determined in continental and marine settings, e.g., during scientific drilling projects conducted in the global framework of the International Ocean Drilling Program (IODP) and the International Continental Scientific Drilling Program (ICDP).

Based on past and present observations, heat-flow data form the basis to decipher the future behaviour of thermal geosystems. Reliable data are a prerequisite for thermal modelling - as boundary condition, validation or calibration input - of various geoenergy aspects (resource assessment and exploitation of geothermal or hydrocarbon systems, long-time and safe geological storage of energy and waste, management and prevention of subsurface utilization conflicts, etc.). In the context of the on-going global energy system transition ('Energie- and Wärmewende'), a well-documented access to parameters and processes of subsurface thermal systems, i.e. the heat flow, is of utmost importance.

Data and Databases - Status Quo – In contrast to the outlined scientific relevance (→ orange box: *What & Why*), the available data and their documentation do not reflect this inherent importance. Until this year, the data structure and documentation was following database concepts that are almost half a century old (cf. *Jessop et al., 1976¹*). Moreover, most of such collections compiled from publications, often pointing back towards rudimentary information of determined heat flow and location coordinates (cf. *Haenel et al., 1988; Hurtig et al., 1992; Hurter & Haenel, 2002*). In preparation for this proposal, we have initiated a global collaboration of heat-flow experts developing a new database schema that includes scientifically relevant data and quality descriptions (*Fuchs et al., 2021a*).

The metadata status is often closely related to the work of individual researchers - usually in a national context - and limited to their period of scientific work (cf. *Cermak, 2007; Cermak et al., 2018*). Consequently, the degree of documentation of heat-flow and metadata is heterogeneous, quite often data values are not even referencing the original data source. Hence, essential metadata for modern data analyses are incomplete or lacking. In **Germany**, for example, the most recent data compilation was published as printed geothermal atlas nearly two decades ago (*Hurter & Schellschmidt, 2003*), considering only c. 60% of the values known today (*Fuchs et al., 2019*). For more than half of the country's area, the heat flow is widely unknown. Where available, it often cannot be easily used as the data quality is not consistently documented (*Fuchs et al., 2019*). Specific heat-flow values mapped in previous compilations reflect different methodical approaches and different levels of detail (→ blue box: *How*), resulting in vast challenges to combine the regional maps to a German Heat Flow Map. Existing **national geoscientific data systems** that compile classical geothermal data, like borehole temperature and thermal rock properties (e.g. the German geothermal information System 'GeotIS' and the professional geophysical information system 'FIS Geophysics'), do neither contain nor aim for incorporating heat-flow data in the future (→ cf. *reference letters*).

On the **global scale**, the International Heat Flow Commission (IHFC), constitutes a natural partner for our project. Founded in 1963, the IHFC was established under the auspices of the IASPEI (International Association of Seismology and Physics of the Earth's Interior) and promoted by

¹ References in italics are external, without italics are from applicants or members of the associated research groups.

other associations of the IUGG. The IHFC collected heat-flow data based on a technological concept with **data filling structures defined in 1975** (Jessop *et al.*, 1976) – long before modern office computer systems, modern database concepts or digital analysis tools were introduced. In the course of time, several concurrent versions and derivatives of the **Global Heat Flow Database** were published and made available on the internet, leading to a confusing situation for the geoscientific community. The **main drawbacks** of using such erratic and insufficiently curated data collections are characterised by (1) the lack of essential metadata; (2) the resulting absence of a deterministic quality control; and (3) the difficult and “user-unfriendly” access (in some cases not even simple queries and search requests are supported) and (4) missing details or references on possibly applied gradual revisions. Assessing the quality of reported heat-flow values and checking their reliability and relevance for subsequent scientific studies is hardly possible in the current situation.

Scientific need – The situation described above hampers scientific research on the Earth’s lithosphere. For geoscientific considerations, misleading geodynamic implications and conclusions may arrive. For industrial applications, it implies the risk of estimating wrong subsurface potentials and resources. This data quality situation is a **matter of debate for years** and the demand to change situation has been pointed out at several levels: in research and conference articles (Artemieva & Mooney, 2001; Hasterok & Chapman, 2011; Lucazeau & Poort, 2015; Fuchs *et al.*, 2019; Lucazeau, 2019) as well as in the on-going discussion of the IHFC (cf. Verdoya, 2019). However, due to the overwhelming workload for any single researcher, only certain aspects were addressed or treated until today and a **systematic collaborative approach on the global scale** is missing so far.

Why is reporting a mere heat-flow value not enough?

The techniques and methods for heat-flow determination have evolved since the first measurement in 1939. Naturally, this took impact on how heat flow has been reported and what kind of additional metadata were documented. The methods applied are adapted to the specific geological setting, possibly implying individual corrections to obtain the real terrestrial heat flow. In many geological settings, it is difficult to distinguish between the conductive terrestrial heat flow *sensu stricto* (→ orange box) and an in-situ heat flow, affected by other overriding conditions and processes (like advection, convection, or heat refraction). To characterize quality and quantity of a specific local heat flow, the documentation of substantial metadata is required. Without the corresponding metadata, the quality of reported heat-flow values cannot be estimated properly. Required metadata comprises, for instance, information on the applied methods, the type, and effect of applied corrections, information on the geographic and tectonic setting, on depth intervals and involved rock types (cf. Fuchs *et al.*, 2021a).

In preparation of this proposal and to learn about the experiences and expectations of the geoscientific community that use heat-flow data in their research, we developed **an online questionnaire sent to hundreds of scientists and published in social media**. The outcome revealed that **the most important requirements and expectations** ranked by the number of mentions are: (1) quality-controlled heat-flow data, (2) consistent storage, (3) easy-to-use map or GIS based user interface as part of a modern data portal (3) comprehensible documentation of data and metadata, (4) introduction of a quality evaluation scheme (community response: n = 53, 04/2020). Most of the survey attendees consider the database in its current form as not ready to be used as a reference, as it forces scientists to go back into the original scientific study (if reported at all) to check for plausibility and essential additional information. The urgent demand for

a revision of the global heat-flow database is further expressed in the **support letters** signed by scientists and representatives of IUGG, IASPEI and IHFC (attached to this proposal).

Our role & current community activities

The GFZ Potsdam actively contributes to the international debate. We are interacting with the heat-flow community in various ways to ensure the transformation of scientific needs into the proposed project. Sven Fuchs and Andrea Förster are members of the IHFC and responsible as IHFC database custodian (S.F.) and as IHFC Secretary (A.F.). In this context, they organized a DFG-funded international meeting on Terrestrial Heat-Flow and the Geothermal Field of the Lithosphere, scheduled in Potsdam (Germany) for May 2020, but actually postponed by two years to 2022 due to the Covid-19 pandemic. With 80+ expected attendees (number registered for 2020), many researchers who contributed to heat-flow research in the past 2–3 decades were expected to attend the workshop. The networking during the initial meeting preparation phase already worked as a nucleus for the proposed collaborative project elements. We were driving this further with a proposal of a 5-year Task Force of the International Lithosphere Program on the global collaborative assessment of data (which was approved on the day of this submission). More than 66 scientists have committed themselves to support actively our work. All our community activities (meetings, workshops, commission work, online questionnaires) serve the idea to identify scientifically relevant data and metadata as well as the demands on information tools and services (including the relevant data formats) that should be included in a new research data infrastructure.

Project goal – With the *World Heat Flow Database Project*, we propose the **development of a new heat-flow research data infrastructure** that offers a modern **web portal** and data repository with quality-assessed, up-to-date, well-documented, extended and restructured heat-flow data including discovery, visualisation, download and data publication options. This portal will be designed as a **standalone reference** and **one-stop shop for heat-flow related data and information**, including information on authors, researchers, publications, and corresponding projects. The service is designed for being interoperable with other geoscientific data services by providing standardised interfaces, data formats, and metadata. Reflecting the FAIR principles, the new database shall provide the basis for the geoscientific community to apply new data science approaches for a better understanding of the geosphere.

Research data infrastructure: On a technical level, we propose to develop an open-access database with a web portal that enables researchers (i) to store data and metadata, (ii) to request DOIs for the data via GFZ Data Services (iii) to visualize data (spatial interpolations, digital thermal boreholes profiles, virtual thermal cross sections and statistical visualizations directly integrated in the web portal), and (iv) to search for data using logical, numerical or visual selection tools (maps) according to a plethora of selectable criteria. This allows users to explore the available data prior to download and without specific software tools.

Collaborative data revision: On the data level, we propose a collaborative revision approach supported by scientists from all continents to revise up to 60% of the global heat-flow data (currently ca. 74.548 datasets from 1403 publications, new release by Fuchs et al., 2021b) within the 3 years of project funding. To target this unique challenge towards an authenticated and curated global database, numerous researchers (n = 66) have committed their support in the framework of a ILP Task Force. These experts will bear the basic workload and act as supervisors for regions with their specific in-depth expertise.

With the publicity of the official project start, we expect to attract more experts. Such activities will also help to trigger a cultural change on how heat-flow data are stored and used in the geoscientific community. This project proposes to communicate, initiate, organize, lead, and moderate this collaborative approach and to transfer the required revision into a long-term research task for the community. The proposed heat-flow research data infrastructure shall support this community effort with corresponding data acquisition, revision, curation, and quality assurance functions.

1.1 State of the art and preliminary work

On the level of **library and information services and web-based data infrastructures**, the **partners GFZ and TUD** are the national developers of modern research data infrastructures with long-standing experience with persistent identifier (PID, e.g. DOI, ORCID, IGSN²) and international metadata standards and strong involvement in NFDI4Earth and the Research Data Alliance. The proposed project will integrate into and benefit from these national and international activities. The GFZ is Germany's national research centre for the solid Earth sciences with the mission to deepen the knowledge of the dynamics of the solid Earth, and to develop solutions for grand challenges facing society. The global heat flow database will be one of these solutions.

1.1.1 GFZ – Working group 'Exploration of thermal geosystems'

The current research activities of the GFZ working group '*Exploration of thermal geosystems*' (part of *Section 4.8: Geoenergy, Department Geosystems*) focus on geothermal exploration, heat-flow and petrophysical analysis, thermal field modelling, georesource assessment, lithosphere thermal studies, borehole temperature logging, and palaeoclimatic analysis. The group has outstanding expertise in thermal-field and heat-flow analysis stemming from several previous research projects and activities (Springer et al., 1998; Förster & Förster, 2000; Norden et al., 2008; Förster & Förster, 2010; Fuchs & Förster, 2010; Norden et al., 2012; Schütz et al., 2014; Schintgen et al., 2015; Fuchs et al., 2015; Fuchs & Balling, 2016a,b; Förster et al., 2016; Norden et al., 2020; Fuchs et al., 2020; Fuchs et al., 2021a,b). The WG worked in subduction related thermal analysis in North and South America (e.g., SFB "Deformationsprozesse in den Anden"), studied the regional thermal conditions of the Arabic plate (e.g., DESERT, DESIRE), and investigated thermal processes across the European plate (e.g., GeoEnergy, INTERREG4A GeoPower, EU-EIT Climate-KIC N-DEMO; DSF Geotermi, NEOGENE). We work on global scale as well as on scales typical for applied geothermics (geothermal energy, storage of energy, carbon dioxide, and nuclear waste). In terms of heat flow, we have screened data in Germany (cf. Norden et al., 2008; Fuchs & Balling, 2016a, Heckenbach et al., 2018; Fuchs et al., 2019; Fuchs & Förster, in prep) and neighbouring countries and paid tribute to the poor data situation by starting a systematic measurement campaign to determine new heat-flow data in Germany more than ten years ago (cf. references at top of section). This progresses until today and will be continued in future. It gives us excellent insights into the data situation and the resulting scientific demands. The WG provides two elected members of the IHFC (as database custodian and Secretary), who revitalized and shaped the on-going international debate about the need and requirement of the proposed project (→ grey box).

1.1.2 GFZ – Library and Information Services

For the past 20 years, the GFZ Library and Information Services (LIS) have been actively involved in various national and international networks, projects and initiatives related to research data management, data publication, persistent identifier (PID) and open science. For example, the

² DOI: Digital Object Identifier, ORCID: Open Researcher and Contributor ID, IGSN: International Geo Sample Number

“invention” of the [DOI](#) for data STD-DOI (Klump et al., 2006), the [Research Data Alliance](#), [World Data System](#), [CODATA](#), [re3data](#) - Registry of Research Data Repositories (e.g. Pampel et al., 2013; Kindling et al., 2017), [COPDESS](#), Enabling FAIR Data in the Earth, space and environmental sciences (Stall et al., 2018). With GFZ Data Services within LIS, a strong infrastructural unit was built as a key player in research data management (Ulbricht et al., 2016; Elger et al., 2018, Conze et al., 2017). [GFZ Data Services](#) is an international research data repository for the Geosciences domain with the first DOI assigned in 2004. In addition to their focus on data from “long tail communities”, they have developed several DOI minting services for larger projects, international services and networks. Among others, GFZ Data Services is data publisher for the several services of the International Association for Geodesy (ICGEM, IGETS, ISG), the World Stress Map, INTERMAGNET, GEOFON, the Geophysical Instrument Pool Potsdam GIPP, TERENO, EnMAP, the Potsdam Institute for Climate Impact Research PIK, the Specialised Information Service for Geosciences FID GEO (Achterberg et al., 2018). GFZ Data Services is responsible for the GFZ activities for the International Geo Sample Number (IGSN minting services and catalogue, central Handle server for c. 10 Mio IGSN in May 2021) (Conze et al., 2017). Beyond these, we have long-standing experiences in the development of technical solutions for databases, registration of PIDs and data portals at GFZ (e.g. SPP Earth Shape, SPP 4DMB, Medusa).

1.1.3 TUD – Geoinformatics

The research activities of the chair of Geoinformatics of TU Dresden (TUD-GI) focus on geoinformation systems and web-based information and research data infrastructures, smart environmental monitoring and modelling, and distributed spatial analysis. TUD-GI has outstanding expertise in the field of research data management and infrastructures (Bernard et al., 2014). TUD-GI coordinated the development of a scientific GDI in the [GLUES project](#), the coordination project of the BMBF measure on sustainable land management. As the common data and service infrastructure, the GLUES GDI facilitates interdisciplinary data and method exchange between the involved scientists (Eppink, 2012). MetaViz (Henzen et al., 2014) has been developed as a tool to allow for exploring metadata information and in particular the visualization of provenance information of simulated data (Henzen et al., 2013). In the projects [COLABIS](#) (BMBF) and [EXTRUSO](#) (EU-ESF) TUD-GI was responsible for the development of web-based information infrastructures to facilitate scientific analysis, a continuous research data management, and the exchange and collaboration of different scientific communities (Henzen, 2016, Wiemann et al., 2018). Since 2019, TUD-GI coordinates the research project [GeoKur](#) (BMBF) GeoKur aims to support the curation and quality assurance already during the research process. Research focus is on the automated metadata acquisition, detailing metadata descriptions including links to ontologies and vocabularies as well as metadata and data quality assurance. In most of the research projects of TUD-GI geoportals or GI (Web-) applications are developed. Beside the conceptual work and the technical implementation, a significant focus is also put on the usability of these applications. TUD-GI has extensive experiences in organizing usability tests and evaluations (Henzen, 2018a,b) and in integrating the results of these evaluations in the development process. TUD-GI builds upon well-established links to national and international networks and initiatives (e.g. AGILE, iEMSS, GDI-DE, INSPIRE, GEOSS, EuroSDR, OGC) addressing not only research communities but also standardization and legislation bodies, various administrative bodies, further stakeholders and industry. Lars Bernard (chair of TUD-GI) is a member of the RFII (*‘Rat für Informationsinfrastrukturen’*), established by the German Joint Science Conference (*‘Gemeinsame Wissenschaftskonferenz’*).

1.2 Project-related publications

1.2.1 Articles published by outlets with scientific quality assurance, book publications, and works accepted for publication but not yet published.

Articles are numbered for each of the applying authors (E = Elger, F = Fuchs, M = Mäs).

- E1 Pampel, H., Elger, K. (2021): 5.6 Publikation und Zitierung von digitalen Forschungsdaten. - In: Putnings, M., Neuroth, H., Neumann, J. (Eds.), *Praxishandbuch Forschungsdatenmanagement*, Berlin, Munich, Boston : Walter de Gruyter, 531-536.
- E2 Elger, K., Bertelmann, R., Ulbricht, D., Haberland, C., Lauterjung, J., Radosavljevic, B. (2018): FAIR am GFZ. - *System Erde*, 8, 1, 52-59. [doi: 10.2312/GFZ.syserde.08.01.8](https://doi.org/10.2312/GFZ.syserde.08.01.8)
- E3 Stall, S., Yarmey, L. R., ..., Elger, K., et al. (2018): Advancing FAIR data in Earth, space, and environmental science. - *Eos, Earth and Space Science News*, 99. [doi: 10.1029/2018EO109301](https://doi.org/10.1029/2018EO109301)
- E4 Conze, R., Lorenz, H., Ulbricht, D., Elger, K., Gorgas, T. (2017): Utilizing the International Geo Sample Number Concept in Continental Scientific Drilling During ICDP Expedition COSC-1. - *Data Science Journal*, 16, 1, 1-8. [doi: 10.5334/dsj-2017-002](https://doi.org/10.5334/dsj-2017-002)
- E5 Ulbricht, D., Elger, K., Bertelmann, R., Klump, J. PanMetaDocs, eSciDoc, and DOIDB-An infrastructure for the curation and publication of file-based datasets for GFZ data services (2016) *ISPRS International Journal of Geo-Information*, 5 (3), 25
- E6 Elger, K., Vincent, W. F., & Barnard, C. (2014). Knowledge capture and data management. In E. Topp-Jørgensen (Ed.), *Management planning for Arctic and northern alpine research stations – Examples of good practices*. (pp. 233-254). Aarhus University, Denmark: DCE – Danish Centre for Environment and Energy, [doi: 10.2312/GFZ.LIS.2014.001](https://doi.org/10.2312/GFZ.LIS.2014.001)
- F1 Fuchs, S., Beardsmore, G.; Espinoza-Ojeda, O. M. ; Gola, G.; Gosnold, W.; Harris, R.; Jennings, S.; Liu, S.; Negrete-Aranda, R.; Neumann, F.; Norden, B., Poort, J.; Rajver, D.; Ray, L.; Richards, M.; Smith, J. (2021a) A new data-base structure for the IHFC Global Heat Flow Database. *International Journal of Terrestrial Heat Flow and Applied Geothermics*, 4(1), [doi: 10.31214/ijthfa.v4i1.62](https://doi.org/10.31214/ijthfa.v4i1.62)
- F2 Fuchs, S.; Norden, B.; International Heat Flow Commission (2021): The Global Heat Flow Database: Release 2021. GFZ Data Services. <https://doi.org/10.5880/fidgeo.2021.014>
- F3 Fuchs, S., Balling, N., & Mathiesen, A. (2020). Deep basin temperature and heat-flow field in Denmark – New insights from borehole analysis and 3D geothermal modelling. *Geothermics*, 83, 101722. [doi: 10.1016/j.geothermics.2019.101722](https://doi.org/10.1016/j.geothermics.2019.101722)
- F4 Fuchs, S. (2018). The variability of rock thermal properties in sedimentary basins and the impact on temperature modelling – A Danish example. *Geothermics*, 76, 1–14. [doi: 10.1016/j.geothermics.2018.06.006](https://doi.org/10.1016/j.geothermics.2018.06.006)
- F5 Fuchs, S., Förster, H.-J., Braune, K., Förster, A. (2018). Calculation of thermal conductivity of low-porous, isotropic plutonic rocks of the crust at ambient conditions from modal mineralogy and porosity: a viable alternative for direct measurement? *JGR Solid Earth* 123(10), 8602–8614, [doi: 10.1029/2018JB016287](https://doi.org/10.1029/2018JB016287)
- F6 Fuchs, S., & Balling, N. (2016). Improving the temperature predictions of subsurface thermal models by using high-quality input data. Part 1: Uncertainty analysis of the thermal-conductivity parameterization. *Geothermics*, 64, 42-54. <http://doi.org/10.1016/j.geothermics.2016.04.010>
- F7 Fuchs, S., & Balling, N. (2016). Improving the temperature predictions of subsurface thermal models by using high-quality input data. Part 2: A case study from the Danish-German border region. *Geothermics*, 64, 1-14. <https://doi.org/10.1016/j.geothermics.2016.04.004>
- F8 Fuchs, S., Balling, N., & Förster, A. (2015). Calculation of thermal conductivity, thermal diffusivity and specific heat capacity of sedimentary rocks using petrophysical well logs. *Geophysical Journal International*, 203(3), 1977-2000. <https://doi.org/10.1093/gji/ggv403>
- F9 Fuchs, S., Förster, A. (2010). Rock thermal conductivity of Mesozoic geothermal aquifers in the Northeast German Basin. *Chemie der Erde – Geochemistry* 80(3), 13–22. <http://dx.doi.org/10.1016/j.chemer.2010.05.010>
- M1 Bernard, L.; Mäs, S.; Müller, M.; Henzen, C.; Brauner, J.(2014): Scientific geodata infrastructures: challenges, approaches and directions. *International Journal of Digital Earth* 7(7): 613–633. <https://doi.org/10.1080/17538947.2013.781244>
- M2 Henzen, C.; Mäs, S.; Bernard, L. (2013): Provenance Information in Geodata Infrastructures. *Geographic Information Science at the Heart of Europe*. Vandenbroucke, Bucher, Crompvoets (Eds). Heidelberg, Springer: 131–151. https://doi.org/10.1007/978-3-319-00615-4_8
- M3 Eppink, F.V.; Wernitze, A.; Mäs, S.; Popp, A.; Seppelt, R. (2012): Land management and ecosystem services: How collaborative research programs can support better policies, *GAIA-Ecological Persp. for Science and Society*, (21): 55–63. <https://doi.org/10.14512/gaia.21.1.14>

1.2.2 Other publications

- E7 Reguzzoni, M., Carrion, D., ..., Elger, K., et al (2021): Open access to regional geoid models: the International Service for the Geoid. - *Earth System Science Data*, 13, 4, 1653-1666. [doi:10.5194/essd-13-1653-2021](https://doi.org/10.5194/essd-13-1653-2021)
- E8 Ince, E. S., ..., Elger, K., et al. (2019): ICGEM – 15 years of successful collection and distribution of global gravitational models, associated services and future plans. - *Earth System Science Data*, 11, 647-674, [doi:10.5194/essd-11-647-2019](https://doi.org/10.5194/essd-11-647-2019)
- F10 Fuchs, S., Norden, B., & Förster, A. (2019). Revisiting the quality of the German heat-flow data towards high demands on documentation, reproducibility, and reliability. Paper presented at the *27th IUGG General Assembly, Montreal, Canada*
- M4 Henzen, C.; Mäs, S.; Müller, M.; Bernard, L.; Tressel, H.; Haase, S. (2014): GeoMetaFacet 2.0 - Interaktive nutzerfreundliche Visualisierung von geographischen Metadaten. *Geoinformatik* 2014.
- M5 Henzen, D.; Mäs, S.; et al. (2016): A Scientific Data Management Infrastructure for Environmental Monitoring and Modelling. *Proceedings of the iEMSs Conference 2016*, Toulouse, France.

2. Objectives and work programme

2.1 Anticipated total duration of the project

The proposed project is scheduled for a period of 5 years. We apply for DFG funding for a **duration of 36 months**. After the funding period, the proposed data information system shall be transformed into a permanent operation and service of the GFZ. A proposal for a follow-up project is considered.

2.2 Objectives

The main objectives of the proposed project are:

- to develop a strong heat-flow research data infrastructure that offers a **standalone reference and one-stop shop for comprehensive information on heat-flow related data**, publications, projects, and researchers. The intended data services will also include a DOI minting service for newly submitted heat-flow data in collaboration with GFZ Data Services. DOI-referenced data publications are fully citable research products and provide means to acknowledge researchers for contributing to the new database. Additional information tools, like the Heat-Flow-Author's Database or the Heat-Flow-Literature Database, will benefit from attributing ORCID's to authors, DOI's to heat-flow-values reporting publications and IGSNs to sample (when available).
- to provide a reassessed, enriched and updated global heat-flow database will improve the scientific usability of global heat-flow data. These are presented in a new data portal and enriched with interactive services for data exploration, visualization, and analysis like interpolation, digital boreholes or spatial profiles.
- to establish a user-oriented data information system following the FAIR and Open principles that is technically interoperable with other geoscientific data services (e.g. EPOS, GeotIs), embedded into the NFDI4Earth activities, and that is interconnected to globally used persistent identifiers, like DOI, ORCID, IGSN, ROR. The system shall allow to publish new heat-flow datasets but also to integrate historic datasets with enriched metadata following international standards (e.g. ISO19115, DataCite, OGC) in machine-readable exchange formats (JSON, XML).
- to provide both, an easy-to-use and fundamental scientific service of quality-proofed and curated heat-flow data (allowing also comments and annotation of historical data) to a wide range of geoscientific research and applications, namely geodynamic and tectonic research, thermal geophysics, and subsurface technologies like geothermal applications, hydrocarbon exploration and underground storage.

Maximizing the usability and benefit for geoscientists and minimizing the effort for scientists during data ingest is the overall guideline for the development of the proposed infrastructure.

2.3 Work programme and proposed research methods

2.3.1 Work packages and research steps

WP Number	WP1	Start–End: M1–M36
WP Title	Requirement analysis, data management & collaborative enrichment	
Lead	GFZ (Sven Fuchs); involved: TUD, collaborative network	
Objectives: To optimize and update the present database towards the new structure and metadata model using the international collaborative network.		

Description of Work: Based on our preparatory work (definition of new data requirements: Fuchs et al., 2021a, data compilation of status quo: Fuchs et al., 2021b) and the fostered interactions with the community (numerous online workshops, ILP Task Force; preparation of Global Collaborative Data Assessment), a new database model will be developed considering the data and metadata. The metadata schema includes metadata for data discovery and contextual metadata and will be based on international standards. It will include, among others, geo-location fields and persistent identifier (WP2) to literature references (DOI), author identification (ORCID), institute identifier (ROR), sample identifier IGSN) and methodological catalogues. The data migrated to this structure will be screened and reviewed before being published. The missing field entries, metadata, and reference details will be updated. Linked-open data vocabularies/ ontologies used in the geoscience community will enrich the metadata and enable search facets.

We will use the first project workshop to develop and finalize new schemes of data classification and quality. Those classes will help to categorize the data and will be used for faceted search and selection in the graphical user interface (WP3). The extended and newly structured data compilation will then be enriched with additional, yet missing information, e.g. quality relevant information like methods used, corrections applied. For this large and protracted task, the primary and secondary literature of each data entry will be screened to identify and take over missing information. Flags on the data quality and status of data information and additional selector variables will be defined and added.

With respect to the importance of the data assessment, 66 researchers (cf. Section 5.3) have committed them self in the ILP Task Force supporting our proposal to address this massive workload in a large international collaboration. Details to the organization are given in WP4. Beside the organization of the collaborating approach, our evaluation work will start with the pre-analysed heat-flow data set of Germany and surrounding countries. Well documented frequent releases (starting with annual frequency) of the database will be provided for download and shared in international trusted data repositories.

Tasks

- Task 1.1 – Inventory of present database and definition of data storage demand (GFZ)
- Task 1.2 – Analysis of the community expectations and requirements (GFZ)
- Task 1.3 – Develop database model and metadata schema (GFZ & TUD)
- Task 1.4 – Implementation of the international collaborative community approach (GFZ)
- Task 1.5 – Development of quality classes and data classification (GFZ + collaborative network)
- Task 1.6 – Data transfer and metadata enrichment, including setup of reference and author sub database with persistent identifiers (GFZ + collaborative network)
- Task 1.7 – Update and extension of historical data (GFZ + collaborative network)

Description of Deliverables

- D1.1 – New database structure (M3), D1.2 – New heat-flow quality classification scheme (M9)
- D1.3 – 50% of initial global dataset updated (M30)

WP Number	WP2	Start–End: M1–M36
WP Title	Data portal back end, architecture & interoperability	
Lead	GFZ (Kirsten Elger), involved: TUD	
Objectives (1) to develop the technical backbone of a web-based data portal, with (2) a doi-minting service and (3) interoperability to other geoscientific data services.		

Description of Work: Based on the community needs, the new database model ([WP1](#)) will be implemented and the architecture of the research data infrastructure will be conceptualized and designed, including (i) the front-end and back-end communication interfaces, (ii) the interaction strategy between front-end and back-end (interfacing), and (iii) the conceptual workflow of the data submission system. The database will be programmed based on experiences and systems available for years at the GFZ. Due to the speed of technological developments, the decision for a SQL-based (relational PostgreSQL, like for the new [World Stress Map](#), also with DOI-minting service in collaboration with [GFZ Data Services](#)) or noSQL-based (graph-based, like for the [Varved Sediment Database](#); Ramisch et al., 2020) **database infrastructure**, will be drawn at the beginning of the project. Both technologies provide open source solutions that fully meet the required capabilities for this project. Here, we can benefit from various in-house experiences.

[WP2](#) focuses on the technical backbone of functions developed in [WP3](#). These functions include a web-based data portal, a project webpage, community areas, a data submission system, an interconnected literature and author database, and a graphical user interface to the database. For user-submitted data with an adequate metadata description, the database infrastructure will allow a **direct link to GFZ Data Services** (via a standardised API) **that initiates the data publication workflow with DOI**. This option, to publish heat-flow data in citable form, is a novelty for heat-flow data and a major incentive for the community to integrate their data in the new infrastructure. The metadata capture will be supported by an online metadata editor, similar to the [GFZ Metadata Editor](#), that assists researchers in providing their metadata in structured form (cf. [WP1](#)) following international standards (e.g. ISO19115, DataCite, OGC) and converts them to machine-readable metadata in JSON and XML formats.

To always **ensure the integrity of the database**, it will be equipped with an authorization and authentication system with different levels of access according to the purpose (e.g. administrator, super user, reviewer, user) and an **internal review area** for newly submitted and revised data before they are published in the productive database. Changes of the data collection will be tracked and publicly documented with each new release. With the new infrastructure, each heat-flow data is connected to the original literature reference (heat flow literature database), its DOI number (where available) and to IGSNs of originating samples (where available). The authors submitting heat-flow data are asked for their ORCID and/or ResearcherID to facilitate later connection to the ORCID records via DOI-referenced data publications. Petrophysical data (e.g. thermal conductivity values or radiogenic heat production values) determined on rock samples will be connected to the sample's IGSN. References, identifiers and researchers will be made available as selector variables in the graphical user interface ([WP3](#)).

Tasks

[Task 2.1](#) – Allocation of physical storage and server capacity ([GFZ](#))

[Task 2.2](#) – Develop an interaction strategy for the front-end-back-end communication structure and the interfacing ([GFZ & TUD](#))

[Task 2.3](#) – Development and programming of database, incl. user authentication system ([GFZ](#))

[Task 2.4](#) – Development of API for metadata exchange, e.g. for data publications via GFZ Data Services, author identification via ORCID and interoperability of with IGSN metadata ([GFZ](#))

[Task 2.5](#) – Conceptual workflow design of data submission system, user interface and back end communication structure ([GFZ](#))

[Task 2.6](#) – Setup of a web-based data portal and programming of a web interface for data and metadata submission process (incl. server-side validity check) and feedback form. ([GFZ](#))

Task 2.7 – Conceptual design and programming data download interface in various file formats or full database dump (GFZ & TUD)

Description of Deliverables

[D2.1](#) – Database (M12+) ; [D2.2](#) – Submission system with technical validity check (M18) ; [D2.3](#) – Revision and user rights system (M18); [D2.4](#) – Commentary and annotation system of heat flow data (M24); [D2.5](#) – Download and data science UI (M32); [D2.6](#) – Data portal with implemented visualization modules from WP3 (M34)

WP Number	WP3	Start–End: M1–M36
WP Title	Data portal front end, exploration and visualization & mapping	
Lead	TUD (Stephan Mäs), involved: GFZ	
Objectives are (1) to develop a web-based front-end for the data portal, with integrated (2) data selection criteria and search functions, (3) visualizations supporting data exploration as well as (4) statistical analysis and interpolation functionalities.		
Description of Work: A web-based data portal will be set up offering a project webpage, community areas, a data submission system, an interconnected literature and author database, and a graphical user interface to the database. The integrated search function will allow faceted search based on defined categories as well as search functions based on keywords, relevant criteria (e.g., type of heat flow, quality criteria, errors), spatio-temporal extent, authors and years etc. In several implementation steps mapping and visualization tools will be developed. These tools support a spatial and/or statistical analysis and to explore and visually analyse the data without downloading and importing data in any other system. The visualizations will include, for example, spatial interpolations (spherical harmonic analysis), descriptive statistical visualizations with uncertainties and the visualization of digital boreholes (1D thermal profiles) and spatial crustal profiles (2D thermal cross-sections; e.g. from mid-oceanic ridge). The submission system will allow for the submission of data in two ways: (a) manually via an input form or (b) automatically via a data file upload function. The upload will be based on a predefined structure for a csv, json or XML file or as an Excel file with a predefined table structure and restricted input options. For this, the most suitable solution will be discussed and selected with the users. The uploaded data and the provided metadata and data descriptions will be automatically validated for logical consistency, completeness and accuracy requirements, as far as technically possible. This validation will be an integrated part of the submission system and is the basis for the attached DOI service (WP2). In case the submitted data fails the validation, the data provider will receive a report enabling him to revise the data submission. The data download and export will be based on standardized interfaces (e.g WMS for map visualizations of interpolations and their uncertainties), or common formats like csv, GeoJson, GeoTIFF or netCDF. The data exploration will be supported by visualizations of digital boreholes and crustal cross-section (e.g. profiles from the Mid-Atlantic Ridge) and allow the comparison to uploaded surface heat-flow raster from modelling studies. The export formats will be discussed with the user communities and established formats will be supported. For Metadata an export conform to the ISO 19115 and 19139 will be implemented. The data portal will also include a feedback form, which allows users to add commentaries and annotations to data already available in the database. This will help to collect observations and experiences from researchers worldwide that are attributed to specific subsets of data. A FAQ and help section will be developed and provide central support information. This section will contain structured contents (for instance as a decision tree) to interactively assist the users		

in all matters or relevance to the heat-flow database. In case users do not find sufficient information to answer their questions, contact information to a helpdesk will be provided. To gain feedback from the user community, the implementation process will be supported by regular **usability evaluations and user studies**. This will assure that the portal and the provided functionality meets the user requirements.

Tasks

Task 3.1 – Conceptual design of user interface, user interactions, visualization, analysis and interpolation functions and back end communication structure (TUD)

Task 3.2 – Development and implementation of a basic web framework and data portal that integrates the functions developed in the following tasks (TUD)

Task 3.3 – keyword based and area based search and selection functions in the GUI to specify selection criteria for data view and download (GFZ & TUD)

Task 3.4 – basic 2D visualization of point data sets based on user selection criteria (TUD)

Task 3.5 – Implementation of regionalized interpolation algorithms (TUD)

Task 3.6 – Statistical analysis and chart visualization of selected data, spatial areas or cross sections

Task 3.7 – Data exploration visualizations (digital boreholes and cross sections)

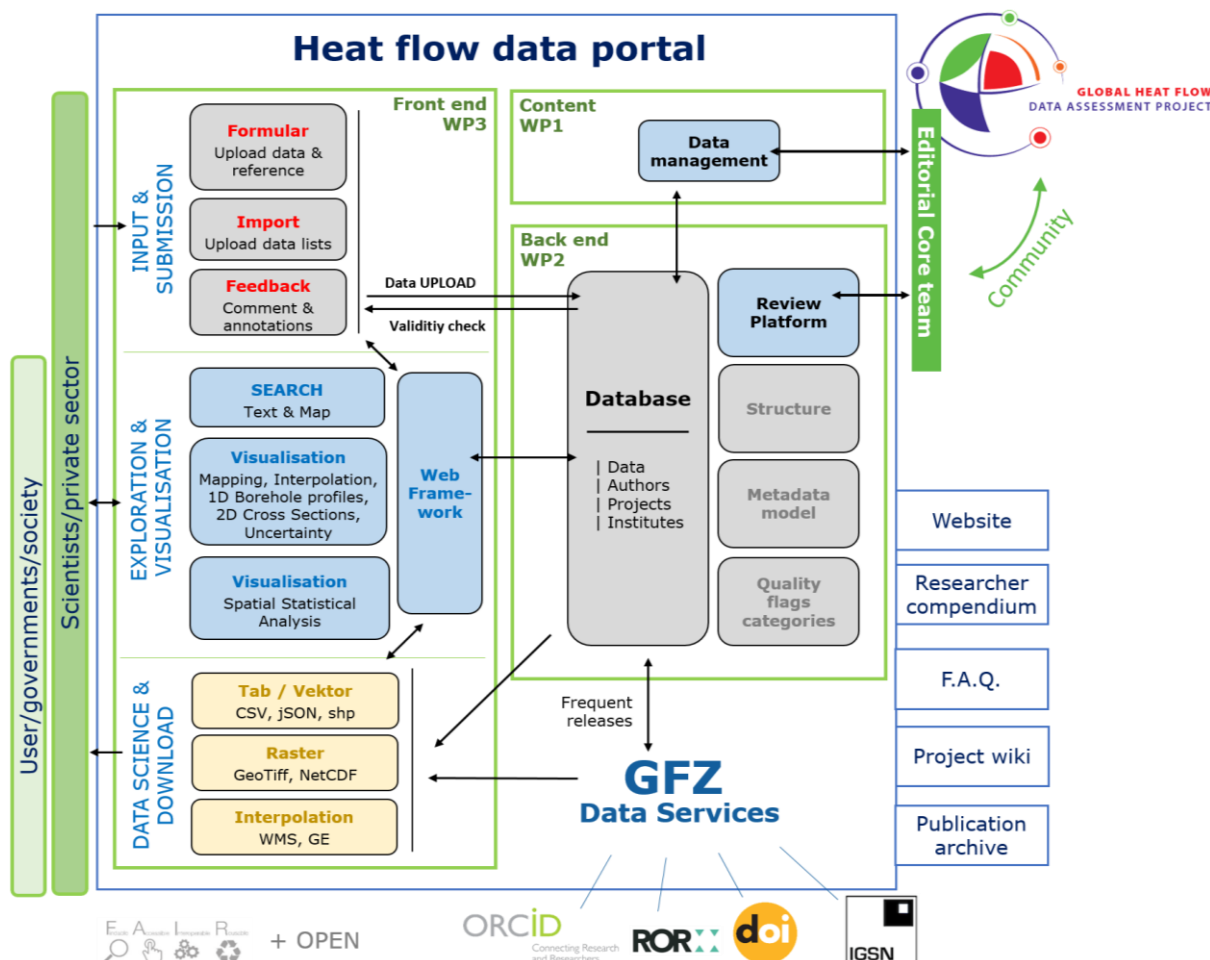
Task 3.8 – Community feedback and usability evaluation (GFZ & TUD)

Description of Deliverables

D3.1 – Basic portal and frontend (M12); **D3.2** – Search functions implemented (M18); **D3.3** – Visualization system (2D point data) with database (M24) ; **D3.4** – Data interpolation functionalities implemented and integrated into the portal (M30); **D3.5** – Statistical analysis and data exploration implemented and integrated into the portal (M36)

WP Number	WP4	Start–End: M1–M36
WP Title	Community, Dissemination, Outreach & Project Management	
Lead	GFZ (Sven Fuchs), involved: TUD	
Description: This WP is aimed to ensure (i) coordination, execution and follow-up of the scientific program (WP1–3) (ii) recruitment, and (iii) management of the timetable and financial resources allocated to the project. It also aims at (iv) coordinating and implementing the collaborative work of the international community, (v) coordinating different communications, (vi) ensuring the widest dissemination and outreach events and (vii) an adequate exploitation of the project results to the scientific community, the public and the stakeholders. It shall maximize the impact on the scientific knowledge in general geoscientific research and development and increase the public awareness and visibility of the project research. For each of these activities, tools, proven workflows and support of administrative staff from GFZ can be used.		
Tasks and Role of Project Partners <u>Task 4.1</u> – Coordination of the consortium agreement (GFZ) <u>Task 4.2</u> – Recruitment (GFZ & TUD) <u>Task 4.3</u> – Implementation of collaborative community approach; org. of continental workshops (GFZ) <u>Task 4.4</u> – Dissemination of scientific results (GFZ & TUD) <u>Task 4.5</u> – Communication and public engagement (GFZ) <u>Task 4.6</u> – Monitoring and controlling of project progress (GFZ) <u>Task 4.7</u> – Coordination of meetings & reports (GFZ & TUD)		
Description of Deliverables D4.1 – Consortium agreement (M1); D4.2 – Recruitment finished (M1); D4.3 – Website and project wiki online (M3); D4.4 – Project newsletter implemented and brochure developed and printed (M6+) D4.5–4.8 – Workshops organized and hold (M9–36); D4.9 – Scientific conferences attended (M12+) D4.10 – Periodic and final reports (M18–36)		

2.3.2 Structural overview



2.3.3 Risks and risk mitigation

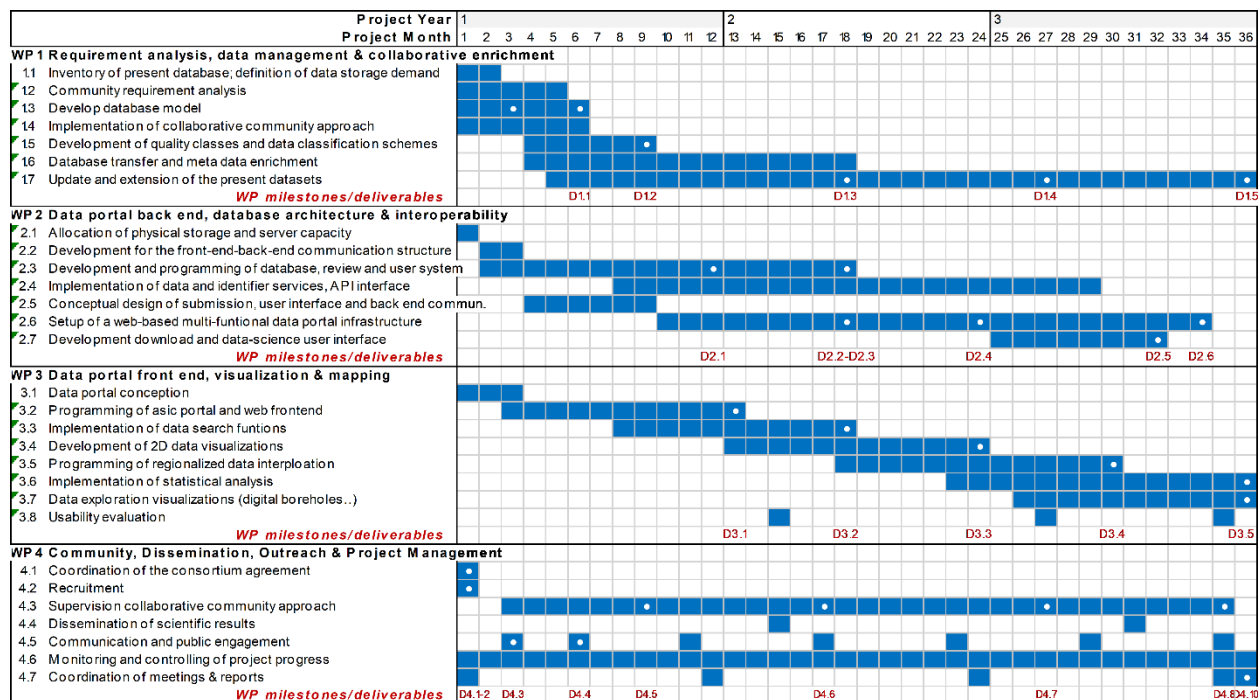
For none of the planned work packages and technical tasks a significant risk is expected. Most planned developments will benefit from rich experiences gained in former projects. Moderate risks are given for the timely recruiting of staff with appropriate programming skills. Depending on the applicant situation, currently fixed-term contracted researcher with the required skills from the partner institutes could be offered to join the project. Unknown risks are given in achieving the deliverable D1.3, which includes the collaborative reassessment of the global heat-flow data. Although we have experiences with the German revision (time aspects) and commitments from the IHFC and many heat-flow experts to assess data over the next 5 years, the offered support is finally purely voluntary and cannot be planned in detail or expected as mandatory for this project. Frequent and open communications, the building of a community spirit and the benefit from common publications of intermediate results shall help to motivate researchers to conduct the required work in time.

2.3.1 Monitoring of project progress

The regular scheduled biweekly telephone conferences and bimonthly status meetings are an important steering tool to ensure the project progress. The iterative data revision of the international heat-flow data, performed by local and regional domain experts (cf. Section 5.3), will be measured by frequent quantitative statistics on the number of revised and updated datasets. For a consistent planning of the work, statistics on the ongoing work will be calculated by the project

lead quartile-wise in dialogue with all committed international supporters. This will build the foundation for continuous adjustments towards the focus shift in the collaborative project part.

2.3.2 Gantt chart with project workflow



2.3.3 Outreach & General public relations work

We will use different communication channels to transport the mission and the project progress towards the public and the science community. The main channel for dissemination toward network members and external users will be the project web portal. This information hub will be actively advertised on external websites and portals supporting a widespread dissemination of the project results. The web portal will include: (i) a website overview of the project, its members and objectives (for experts and the general public); (ii) a public area where science publications, outreach material, announcements of workshops and meetings, shared data/software and other resources will be made available to the science community; (iii) a restricted area for sharing documents, data and tools within the consortium. The web portal will also host a wiki and a blog where short news on recent developments and activities are provided and informal exchanges between project and scientific community are fostered. The maintenance of the web site will be realized by the GFZ IT under the responsibility of the management team, the project lead and the responsible work package leaders.

The project newsletter - annually published by the GFZ - will provide a proactive way to communicate the project activities to members and the public complementary to the website. A flyer/brochure will be produced, written in an easy-to-understand language, for distribution in local events, conferences, and public meetings. Beside web portal and newsletter, social media channels will be used to inform the scientific and general public about our activities and results. Major channels that will be used to disseminate announcements, material presented in workshops and research results are: YouTube, ResearchGate, LinkedIn, Twitter and similar standard online platforms. We will interact with national and international stakeholders and entities (e.g. IASPEI, IUGG, EGA, EGE) to maximize public awareness.

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4. Supplementary information on the research context

4.1 General ethical aspects

- not applicable –

4.2 Governance

The proposed project formed by GFZ and TU Dresden interacts with numerous external colleagues and national and international partners. The applicants hold the database custodianship of the International Heat Flow Commission (IHFC) as well as lead and organize a group of international experts (currently 66 scientists), committed to support the data revision part of our project. This experts group will formally constitute a 5-year Task Force of the International Lithosphere Program (ILP), funded by IUGG and IUGS (networking seed funding) that will bear the enormous workload of the global data assessment. Selected IHFC members and ILP-TF leaders will build an advisory board and core team, both to support the communication between community and project as well as to check and prove the legwork of the community. The overall project will integrate into the developments of the German National Research Data Infrastructure and, in particular, integrate and support into the framework provided by NFDI4Earth.

4.3 Measures to meet funding requirements and handle project results

The GFZ Potsdam has operated open-science data services for years and has gained experiences in developing and implementing these community services successfully. Similar to the proposed project, re3data.org stands as one example of service that has been developed in Germany and is nowadays an internationally recognized scientific resource and web portal. Initially started as a third-party-funded project, it became part of the regular and permanent operations of the GFZ to ensure their sustainability. A similar long-term sustainability concept is developed for the proposed nation-wide heat-flow data infrastructure. Concrete actions that will be taken after funding has expired are listed in 2.4.1. All project partners support the networking and interconnection of the project within the community. All the findings of the project will not be only **announced to the relevant geoscientific community** (cf. WP4), but will be **made also available to third parties free of charge**. As we use open-access software packages for the programming work, source code produced will be made available **open source**.

4.3.1 Sustainability plan

With the presented concept, we propose an initial financing for a long-term nation-wide research data structure of international importance. The **GFZ**, as one of the largest geoscientific research centres in Europe, is able and **committed to maintain and safeguard** the project findings and ensure their sustainability. The system will be driven and maintained by the Section Geoenergy (professional and technical part) with support of the IT and the LIS section (administrative and

technical part) of the GFZ. A **permanent position** at the Geoenergy section will sustain the long-term professional maintenance of the database after funding expired. The broad and long-term experience of the members of the LIS Section will **ensure** that **any relevant technical standards** particularly those concerning data collection and data storage are considered during establishment and enhancement of the research data infrastructure. This infrastructure is **designed for the needs** of the community, measured by thorough analysis of several surveys conducted in the run-up of this proposal (cf. Starting point). These **requirements will be monitored** in future by collecting feedback and continuing our activities in national (German Geophysical Society DGG) and international boards (International Heat Flow Commission IHFC). Based on this monitoring we will adapt our activities if required. Core elements of the new service, like the interoperability with DOI, ORCID and IGSN systems, are ensured as these data services are **already implemented** in the permanent structure of **GFZ Data Services**. The new data portal will be developed in accordance with the work of the NFDI4Earth and will be implemented there.

4.3.2 Working database and repository selection

The proposed data portal is designed as a living scientific information service with an actively maintained and processed working database. This requires the development of a technical infrastructure that specifically serves this goal and enables advanced subject-specific visualisations as well as special data analysis tools and models. The advancement of stationary repositories and data archives for this purpose, like PANGAEA and others, which are originally designed for the long-term storage and archiving of research data does not serve the outlined idea. Despite the broad acceptance and support of our community, the main asset of this project will be the curated database. We will frequently release the database as open-access data publication (with GFZ Data Services). We will also support the long-term storage of these releases in any interested repository. Representatives of Pangaea, where the 2012 version of the IHFC database was achieved, were contacted and informed about this proposal (March 2021).

4.4 Formal assurances

Wir erklären, die Regeln guter wissenschaftlicher Praxis einzuhalten gemäß dem DFG-Merkblatt „Werkzeuge und Verfahren des wissenschaftlichen Informationsmanagements“ und der DFG- Denkschrift „Sicherung guter wissenschaftlicher Praxis“. Bei Förderung verpflichten wir uns, die bewilligten Mittel ausschließlich im Interesse einer zielstrebigem Verwirklichung des geförderten Vorhabens einzusetzen. Bei der Verwendung und Abrechnung werden die einschlägigen Richtlinien der DFG beachtet. Darüber hinaus verpflichten wir uns, der DFG zu den im Bewilligungsschreiben angegebenen Terminen über den Fortgang der Arbeiten zu berichten und Nachweise über die Verwendung der Beihilfe vorzulegen. Wir bestätigen, dass die aus dem Projekt resultierenden Publikationen ebenso wie einschlägige Dokumentationen im Open Access verfügbar gemacht und Dritten zur umfassenden Nachnutzung bereitgestellt werden. Wir erklären außerdem, dass der Quellcode der im Projekt entwickelten Software nach den Prinzipien von Open Source dokumentiert und für die Nachnutzung durch Dritte verfügbar gemacht wird.

4.5 Information on scientific and financial involvement of international cooperation partners

The collaborative part of the proposed project will be conducted in **close cooperation with an international network** of heat-flow experienced colleagues, among them the members of the International Heat Flow Commission (IHFC) of the International Association of Seismology and Physics of the Earth's Interior (IASPEI). The IHFC members, currently 23 researchers from 5 continents with long experience in heat-flow research, will constitute the backbone of geoscientists that bear the international collaborative revision of the database. Another 43 colleagues committed themselves in the frame of the ILP Task Force proposal. **None of them applied for individual funding** related to our proposal yet. The ILP task force includes an annual seed funding for networking costs of 4000€. Although the proposed project is **not set up as an official** part of ongoing **agreements** between the DFG and other partner organisations, some of the collaborating researchers are located at such institutes. This include researchers from institutes like the Centre National de la Recherche Scientifique (CNRS, FR), the Consiglio Nazionale delle Ricerche

(CNR, IT), the Czech Academy of Sciences (CAS, CZ), and the Russian Academy of Sciences (RAS, RU). IHFC and TF members will act as a multiplier towards the broader geoscientific community, related exploration industry, and other stakeholders to maximize the public awareness of this project. Amongst others, **support letters** of the IHFC, the IASPEI and the IUGG are attached to this proposal.

5. People/collaborations/funding

5.1 Employment status information

All partners are **first-time DFG applicants**.

GFZ: Fuchs, Sven Dr. Postdoc, fixed term contract, 03/2022, Helmholtz Society & EU EFRE
Elger, Kirsten Dr. Scientist, fixed term contract, 12/2022, GFZ budget
TUD: Mäs, Stephan Dr. Scientist, permanent contract

5.2 Composition of the project group (budget funds)

GFZ: Norden, Ben Dr. Scientist, Section Geoenergy – permanent contract, 30%
Ott, Florian Library and Information Services, GFZ
Förster, Andrea Dr. Scientist, Section Geoenergy – guest scientist
Schöbel, Birgit Scientist, Section Geoenergy – permanent contract, 30%
TUD: Bernard, Lars Prof. Chair of Geoinformatics – permanent contract

5.3 Institutions or researchers in Germany with which/whom you have agreed to cooperate on this project

Dr. Ricarda Dziadek (AWI), Dr. Christophe Galerne (Universität Bremen), Dr. Almudena García-García (Helmholtz Centre for Environmental Research - UFZ), Dr. Norbert Kaul (University of Bremen), Prof. Thomas Kohl (Karlsruher Institut für Technologie), Prof. Christian Müller (Fielax), Prof. Magdalena Scheck-Wenderoth (GFZ Potsdam)

5.4 Institutions or researchers abroad with which/whom you have agreed to cooperate on this project

*Scientists that committed to support the collaborative revision of the database (*TF Co-Leader):*

Global: International Heat Flow Commission (IHFC) of the IASPEI, Task Force of the International Lithosphere Program ('Lithospheric Heat Flow – Global Data Assessment Project')

Asia: Dr. Xiang Gao (Institute of Oceanology, Chinese Academy of Science), Dr. Jiang, Guangzheng (State Key Laboratory of Lithospheric Evolution, Chinese Academy), Dr. Lijuan He (Chinese Academy of Sciences), Prof. Shengbiao Hu (Chinese Academy of Sciences), Prof. Shaopeng Huang (Shenzhen University), **Prof. Shaowen Liu*** (Nanjing University), Dr. Xiaobin Shi (South China Sea Institute of Oceanography, Chinese Academy of Sciences)

Australia: **Dr. Graeme Beardsmore*** (University of Melbourne), Mr Ed Gerner (Geoscience Australia), Dr. Derrick Hasterok (University of Adelaide), Mr Marcus Haynes (Geoscience Australia), Dr. Chris Matthews (University of Adelaide), Prof. Sandra McLaren (University of Melbourne), Dr. Tim Rawling (AuScope Ltd), Dr. Joanne Whittaker (U. of Tasmania)

Europe: Dr. Vladimir Zui (Unitary Republic Enterprise), Dr. Vladimir Cermak (Czech Academy of Sciences), Dr. Petr Dedecek (Czech Academy of Sciences), Dr. Jan Safanda (Czech Academy of Sciences), Dr. Irina Artemieva (University of Copenhagen), Prof. Niels Balling (Aarhus University), Dr. Argo Joeleht (University of Tartu), Prof. Ilmo T. Kukkonen (University of Helsinki), Dr. Jeffrey Poort (Sorbonne University), Prof. Frédérique Rolandone (Sorbonne Université, CNRS-INSU, Institut des Sciences de la Terre Paris), Dr. Paolo Chiozzi (University of Genoa), Dr. Gianluca Gola (National Research Council of Italy), **Prof. Massimo Verdoya*** (University of Genoa), Dr. Dmitry Demezhko (Russian Academy of Sciences), Prof. Mikhail Khutorskoy (Russian Academy of Sciences, Moscow, Russia), Dr. Guzel Vakhitova (Bashkir State University), Dr. Dušan Rajver (Geological Survey of Slovenia), Prof. Fidel González-Rouco (Complutense University of Madrid), Dr. Ignacio Marzan (Complutense University of Madrid), Dr. Ana Maria Negredo Moreno (Complutense University of Madrid), Prof. Ladislaus Rybach (ETH Zürich, Institute of Geophysics)

North America: Prof. Hugo Beltrami (St. Francis Xavier University), Dr. Francisco Cuesta-Balero (Memorial University of Newfoundland), Prof. em. Alan M. Jessop (Geological Survey of Canada), Dr. William Colgan (GEUS), Prof. Juan Contreras (Center for Scientific Res. and Higher Edu. at Ensenada), Dr. Orlando Miguel Espinoza-Ojeda (Michoacan University of Saint Nicholas of Hidalgo), **Dr. Raquel Negrete Aranda*** (Center for Scientific Res. and Higher Edu. at

Ensenada), Dr. Florian Neumann (Center for Scientific Res. and Higher Edu. at Ensenada), Prof. William D. Gosnold (University of North Dakota), **Prof. Robert Harris*** (Oregon State University), Dr. Maria Richards (SMU Geothermal Laboratory), Dr. Jared Smith (University of Virginia), Dr. Carol Stein (University of Illinois at Chicago)
South America: Dr. Valiya Hamza (Brazil)

5.5 Institutions or researchers with which/whom you have worked on projects within the past three years

GFZ: *National:* KIT (Thomas Kohl, Thomas Forbriger, Hans-Jürgen Göbelbecker); Institut für Bibliotheks- und Informationswissenschaft (Vivien Petras); Niedersächsische Staats- und Universitätsbibliothek Göttingen (Wolfram Horstmann); TIB - Technische Informationsbibliothek (Britta Dreyer); Uni Göttingen (Bianca Wagner); UFZ (Thomas Schnicke, Jan Bumberger), *International:* Aarhus University (Denmark; Niels Balling); GEUS, Copenhagen (Denmark, Anders Mathiesen, Lars-Henrik Nielsen); Geologische Bundesanstalt (Österreich, Gregor Götzl), Purdue University Libraries (USA, Michael Witt), Université Laval (Québec, Canada: Warwick Vincent, Christine Barnard/ CEN, ArcticNet), British Geological Survey (Keyworth: Helen Graves, Helen Taylor, Edinburgh: Simon Flower/ INTERMAGNET), Utrecht University (The Netherlands: Richard Wessels, Otto Lange), INGV (Rome, Italy: Massimo Cocco, Lily Freda, Aldo Winkler), Università Roma Tre (Rom, Italy: Francesca Funicello), American Geophysical Union (Shelley Stall), CSIC (José-Luis Fernandez-Turiel), ETH Zürich (Alba Zappone, Florian Haslinger), DataCite (Martin Fenner, Patricia Cruz), University of Columbia/ Lamont Doherty Earth Observatory (Kerstin Lehnert/ IEDA)

TUD: *National:* UFZ Leipzig (Ralf Seppelt), HTW Dresden (Frank Schwarzbach), LMU München (Wolfram Mauser, Florian Zabel), iDiv Uni Leipzig (Carsten Mayer), 52°North (Albert Remke, Simon Jirka), HS Beuth Berlin (Petra Sauer), Kiel Earth Institute Christian-Albrechts-Universität zu Kiel (Gernot Klepper, Ruth Delzeit); Potsdam Institut für Klimafolgenforschung (Alexander Popp), *International:* Universität Salzburg (Barbara Hofer), IIT Mandi India (Dericks Shukla), University of Free State South Africa (Marinda Avenant, Maitland Seaman), North West University South Africa (Adeline Ngie)

5.6 Project-relevant cooperation with commercial enterprises

- not applicable -

5.7 Project-relevant participation in commercial enterprises

- not applicable -

5.8 Other submissions

- not applicable -

6. Requested modules/funds

6.1 Requested modules/funding

6.1.1 Basic Module

For each applicant, we apply for funding within the Basic Module.

Applicant: Fuchs, Sven (GFZ)

Category	Applied funding	Description
Personal	Postdoc (n.n.) , 36 person months (PM), 100%	Person for work in WP1 and WP4
Personal	Student worker , 36 PM, 40h/month (sum 30,140 €)	Help with data screening and literature review and assessment, no special knowledge required, Start M1
Travel costs	7,000 €	Covers travel costs for project meetings (2-3 meetings/year; 500 €/year) and visits at conferences and scientific meetings (2000 €/year), like Annual DGG Assembly, EGU 2022, AGU 2022/2024, and IUGG 2023
Guests	4,000 €	Covers visits of min. 4 incoming international guests at the GFZ. Assuming an average duration of 5 days, we estimate the cost of

		one visit at 600 € for traveling and 80 €/night for accommodation, summing up to 1000 € per guest
Publication costs	2,250 €	For payment of publication fees for open-access journals.(750 € / year)

Applicant: Elger, Kirsten (GFZ)

Category	Applied funding	Description
Personal	Scientific Associate , 36 PM, 100%	Person for lead and work in WP3, support of WP2
Personal	Student worker , 36 PM, 40h/month (sum 30,140 €)	Help with literature review, tools & website programming, Start early 2021
Travel costs	6,000 €	Covers travel costs for project meetings (500 €/year) and visits at conferences and scientific meetings (1500 €/year), like DGGV/DGG Assembly, RDA Plenary, IUGG 2023
Publication costs	2,250 €	For payment of publication fees for open-access journals. (750 € / year)

Applicant: Mäs, Stephan (TUD)

Category	Applied funding	Description
Personal	Scientific Associate 36 PM, 100%	Person for lead and work of WP2, support of WP3
Personal	Student worker , 36 PM, 40h/month (sum 30,140 €)	Help with literature review, tool programming and usability evaluation, Start early 2021
Travel costs	6,000 €	Covers travel costs for project meetings (500 €/year) and visits at conferences and scientific meetings (1500 €/year), like RDA Plenaries, EGU, GIScience or AGILE in 2022, 2023, 2025, IUGG/IASPEI 2023
Publication costs	2,250 €	Publication fees for OA journals. (750 € / year)

GFZ and TUD will cover the workspace, computing needs, and consumables for its staff as basic support.

6.1.2 Module: Project-specific workshops

Sven Fuchs applies for funding within the module “project-specific workshops”.

Institution	Appl. fund.	Description
GFZ	9,000 €	Workshop 1: 2-day workshop “Workflow and conceptual design” Costs for travel & accommodation (6,000 €), organization (300 €), catering (1,200 €) and documentation (1,500 €), ca. 10–15 international participants.
GFZ	19,000 €	Workshop 2 3-day workshop “Database revision: half-time’s results and outlook” Costs for travel & accommodation (11,000 €), organization (2,500 €), catering (4,000 €) and documentation (1,500 €), ca. 30 international participants.
GFZ	2,500 €	Workshop 3 1-day workshop “The World Heat Flow Database – Relaunched” Costs for organization (200 €), catering (1,500 €) and documentation (800 €)
<i>Sum without overhead: 30,500 €</i>		

6.2 Financial contributions

The participating institutions provide some personal from the permanent staff (shared time, cf. Section 5.2) to support the work, controlling, and the management of the project. On top, all project partners offer and provide their own infrastructural or financial contributions, like office space, equipment, costs of IT infrastructure and maintenance, and support with controlling and administration. The purchase of large additional equipment is not required and direct funding from GFZ and/or TUD to the project is not provided.